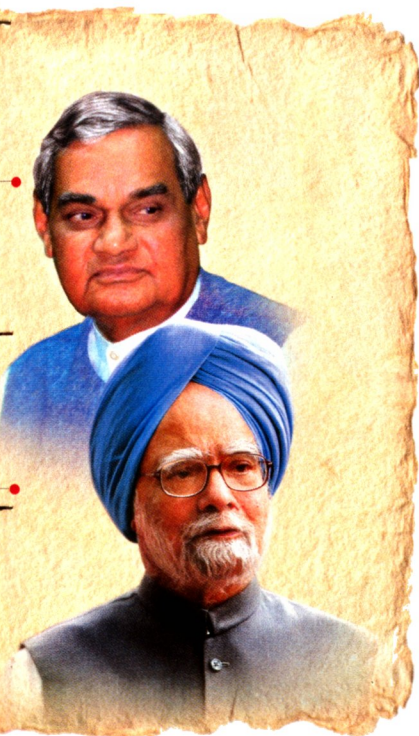




cent Submarine Design Group (SDG) which actually designed the ATV trained with Russia's Rubin design bureau. Funds for the Rs 2,800-crore project were never a problem and were sanctioned from the cabinet secretariat, and the joint DRDO-Navy project was always a closed loop within the Prime Minister's Office. After Indira's assassination, the ATV baton passed on to Rajiv Gandhi who was also the defence minister. "Rajiv understood both technology and strategy and was in favour of the project. He would keenly participate in our discussions on whether our N-submarine needed one reactor or two and the availability of enriched uranium for the reactors," says a former project official.

In January 1988, Rajiv donned work overalls and boarded the INS *Chakra* as she steamed into Visakhapatnam to join the navy. He became the only Indian prime minister to board a nuclear-powered submarine. The return of the *Chakra* at the expiry of its

lease in 1991 coincided with the implosion of the former Soviet Union, the tectonic event that nearly killed the project. Officials say there was a perceptible lack of political interest in the project on both sides: President Boris Yeltsin in Russia and prime minister Narasimha Rao. The SDG, meanwhile, began converting the Charlie-1 designs for industrial manufacture.

The Indian private sector was chosen to build the 104-metre-long prototype, dubbed S2. Larsen & Toubro (L&T) built the hull, Tata Power made the control systems and Walchandnagar Industries made the complex high pressure pumps and valves which carried saturated steam. The BARC had still not succeeded in perfecting the reactor so the Government decided to continue reactor development parallel to the construction of the first submarine.

On January 5, 1998, in a quiet ceremony at the L&T's Hazira facility, the then DRDO chief APJ Abdul Kalam sym-

bolically cut the first steel plate of the ATV. The project picked up speed under the NDA and during the tenure of Atal Bihari Vajpayee, the prime minister who stunned the world by bringing India out of the nuclear closet. Vajpayee, who also headed the newly-established NCA, chaired the apex committee of the ATV (see accompanying graphic).

The project also had two other authorities—the political and the executive council. The project remained under the direct control of Vajpayee through his national security adviser, Brajesh Mishra. Talks for the lease of another nuclear submarine with Russia were revived. In January 2004, India and Russia signed a secret \$650-million intergovernmental agreement (IGA) for the completion and lease of one unfinished Akula class nuclear-powered attack submarine and training crews to man them. (The submarine, also called the *Chakra*, is undergoing trials in Russia and is expected to join the navy later this year). The crucial part of the IGA was the assistance to build the reactor, which had delayed the project by years.

The project entered its last mile during Manmohan Singh's first tenure in 2004. He attended several meetings and would often ask project officials, "Everything alright?" The query was a mere formality because the project received unstinted support. In 2005, the UPA Government gave an in-principle clearance for building a follow-on series of larger ballistic missile submarines (SSBNs), costing nearly Rs 8,000 crore a piece or nearly twice that of the current series of ATVs and another line of nuclear-powered fast attack submarines (SSNs) to escort them. "If you need money, you'll get it," the then finance minister P. Chidambaram had assured the project team. The last and most important milestone was reached in 2006 when an indigenously-built version of the Russian VM-4 PWR, which propelled the Charlie-1, was successfully land-tested and sealed into the hull of the ATV the following year.

As Singh walks towards the *Arihant* on July 26, he can have the satisfaction of having supervised the final chapter in India's nuclear destiny. ■

CHAIN OF COMMAND

The decision-makers who guide the nuke submarine project

